Chapter 10. Inner Product Spaces

Orthonormal bases, orthogonal and unitary groups, adjoints

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10.1 Inner product spaces

Motivation. The four properties of the dot product used in Chapter 9 define an inner product on any real or complex space.

Definition 10.1.1 (Hermitian dot product)

For $X = (a_1, \ldots, a_n)^t$, $Y = (b_1, \ldots, b_n)^t \in \mathbb{C}^n$,

$$\langle X, Y \rangle = X^t \cdot \bar{Y} = \sum_{i=1}^{n} a_i \bar{b}_i$$

is the (**Hermitian**) **dot product** of $\mathbb{C}^n$.

Remark 10.1.2

Without conjugation, $(1+i, 1-i)^t$ would have "length" 0 ($(1+i)^2 + (1-i)^2 = 0$) and $(i, 0)^t$ an imaginary "length".

Exercise 10.1.3

For $X, Y, Z \in \mathbb{C}^n$, $c \in \mathbb{C}$: (a) $\langle X+Y, Z\rangle = \langle X, Z\rangle + \langle Y, Z\rangle$, $\langle cX, Y\rangle = c\langle X, Y\rangle$;
(b) $\langle X, Y+Z\rangle = \langle X, Y\rangle + \langle X, Z\rangle$, $\langle X, cY\rangle = \bar{c}\langle X, Y\rangle$;
(c) $\langle X, Y\rangle = \overline{\langle Y, X\rangle}$;
(d) if $X \neq 0$, then $\langle X, X\rangle \in \mathbb{R}$ and $\langle X, X\rangle > 0$.

Solution. (a), (b) from $\overline{b+c} = \bar{b} + \bar{c}$, $\overline{cb} = \bar{c}\bar{b}$.

(c) $\overline{\sum b_i \bar{a}_i} = \sum \bar{b}_i a_i$.

(d) $\langle X, X\rangle = \sum |a_i|^2$. □

Definition 10.1.4 (Inner product on F^n)

($F = \mathbb{R}$ or $\mathbb{C}$.) A function $(X, Y) \mapsto \langle X, Y\rangle$, $F^n \times F^n \to F$, is an **inner product** on F^n if for all $X, Y, Z \in F^n$, $c \in F$: (1) $\langle X+Y, Z\rangle = \langle X, Z\rangle + \langle Y, Z\rangle$; (2) $\langle cX, Y\rangle = c\langle X, Y\rangle$; (3) $\langle X, Y\rangle = \overline{\langle Y, X\rangle}$; (4) if $X \neq 0$, then $\langle X, X\rangle \in \mathbb{R}$ and $\langle X, X\rangle > 0$. (For $F = \mathbb{C}$, also a **Hermitian inner product**.)

Observation 10.1.5

If $\langle \cdot, \cdot \rangle$ is an inner product on F^n: (a) $\langle X, Y+Z\rangle = \langle X, Y\rangle + \langle X, Z\rangle$;
(b) $\langle X, cY\rangle = \bar{c}\langle X, Y\rangle$.

Proof. By (3), $\langle X, Y+Z\rangle = \overline{\langle Y+Z, X\rangle} = \overline{\langle Y, X\rangle} + \overline{\langle Z, X\rangle} = \langle X, Y\rangle + \langle X, Z\rangle$, and $\langle X, cY\rangle = \overline{c\langle Y, X\rangle} = \bar{c}\,\overline{\langle Y, X\rangle} = \bar{c}\langle X, Y\rangle$. ■

Definition 10.1.6 (Inner product space)

Let V be an F-vector space (possibly infinite-dimensional). $\langle \cdot, \cdot \rangle : V \times V \to F$ is an **inner product** if for all $u, v, w \in V$, $c \in F$: (1) $\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$; (2) $\langle cv, w\rangle = c\langle v, w\rangle$; (3) $\langle v, w\rangle = \overline{\langle w, v\rangle}$; (4) if $v \neq 0$, then $\langle v, v\rangle \in \mathbb{R}$ and $\langle v, v\rangle > 0$ (*positive definiteness*). V with

an inner product is an **inner product space**; for $F = \mathbb{C}$, $\langle \cdot, \cdot \rangle$ is also called **Hermitian**.

Remark 10.1.7

(a) Observation 10.1.5 still holds.
(b) For $F = \mathbb{R}$, conjugation does nothing, so a real inner product is a *symmetric* bilinear form.
(c) For $F = \mathbb{C}$, an inner product is linear in the first variable and **conjugate linear** (anti-linear) in the second.
(d) Every F-subspace of an inner product space is an inner product space.

Exercise 10.1.8

$\langle X, Y \rangle = \sum_{j=1}^{n} j\, a_j b_j$ is an inner product on $\mathbb{R}^n$.

Solution. (1)–(3) are clear; $\langle X, X \rangle = \sum j a_j^2 > 0$ for $X \neq 0$. □

Exercise 10.1.9

$\left\langle \binom{a}{b}, \binom{c}{d} \right\rangle = 2ac - bc - ad + 2bd$ is an inner product on $\mathbb{R}^2$.

Solution. It is bilinear and symmetric, and $\langle (a,b),(a,b) \rangle = 2a^2 - 2ab + 2b^2 = a^2 + b^2 + (a-b)^2 > 0$ for $(a,b) \neq 0$. □

Exercise 10.1.10

If $\langle \cdot, \cdot \rangle_1, \langle \cdot, \cdot \rangle_2$ are inner products on V, so is $\langle v, w \rangle = \langle v, w \rangle_1 + \langle v, w \rangle_2$.

Solution. (1)–(3) are additive; for $v \neq 0$ both summands are positive reals. □

Exercise 10.1.11

$\langle f(t), g(t) \rangle = \int_0^1 f(t) g(t)\, dt$ is an inner product on $\mathbb{R}[t]$.

Solution. Bilinearity and symmetry are clear. If $f \neq 0$, $f^2 \geq 0$ is continuous and positive except at finitely many points, so $\int_0^1 f^2 > 0$. □

For $f = f_1 + i f_2 : \mathbb{R} \to \mathbb{C}$ (real and imaginary parts), $\int_a^b f = \int_a^b f_1 + i \int_a^b f_2$; f is continuous iff f_1, f_2 are.

Exercise 10.1.12

$\langle f(t), g(t) \rangle = \int_0^1 f(t) \overline{g(t)}\, dt$ is an inner product on $\mathbb{C}[t]$.

Solution. As in Exercise 10.1.11, with $\langle f, f \rangle = \int_0^1 |f|^2$ and $\overline{\int f \bar{g}} = \int \bar{f} g$. □

Exercise 10.1.13

$C^0[0, 2\pi] = \{f : [0, 2\pi] \to \mathbb{C} \text{ continuous}\}$ is a $\mathbb{C}$-vector space, and $\langle f, g \rangle = \frac{1}{2\pi} \int_0^{2\pi} f(x) \overline{g(x)}\, dx$ is an inner product on it.

Solution. Sums and multiples of continuous functions are continuous. (1)–(3) as before; if $f \neq 0$, $|f|^2$ is continuous, nonnegative and positive on an interval, so $\langle f, f\rangle > 0$. □

Exercise 10.1.14

$\langle A, B\rangle = \operatorname{tr}(A^t \cdot \bar{B})$ is an inner product on $\mathfrak{M}_{n,n}(F)$ (the **trace form**).

Solution. $\operatorname{tr}(A^t\bar{B}) = \sum_{i,j} A_{ji}\bar{B}_{ji}$ is the dot product of A and B viewed as vectors in F^{n^2}, so Exercise 10.1.3 applies. □

10.2 Properties of inner product spaces

Motivation. Norm, Pythagoras, Cauchy–Schwarz and the triangle inequality hold in any inner product space.

Throughout, V is an inner product space over F.

Exercise 10.2.1

If $\langle w, v\rangle = 0$ for all $w \in V$, then $v = 0$.

Solution. Take $w = v$ and use positive definiteness. □

Definition 10.2.2 (Norm)

$\|v\| = \sqrt{\langle v, v\rangle}$.

Exercise 10.2.3

(a) $\|v\| \geq 0$, with equality iff $v = 0$.
(b) $\|cv\| = |c|\|v\|$.
(c) If $v \neq 0$, $\frac{1}{\|v\|}v$ is a **unit vector**.

Solution. (b) $\langle cv, cv\rangle = c\bar{c}\langle v, v\rangle = |c|^2\langle v, v\rangle$. The rest as in Exercise 9.1.5. □

Exercise 10.2.4 (Parallelogram law)

$\|v + w\|^2 + \|v - w\|^2 = 2(\|v\|^2 + \|w\|^2)$.

Solution. $\|v \pm w\|^2 = \|v\|^2 \pm (\langle v, w\rangle + \langle w, v\rangle) + \|w\|^2$; add. □

Definition 10.2.5 (Perpendicularity)

(a) $v \perp w$ if $\langle v, w\rangle = 0$ (equivalently $\langle w, v\rangle = 0$).
(b) For $S, T \subseteq V$, $S \perp T$ if $\langle v, w\rangle = 0$ for all $v \in S$, $w \in T$.

Exercise 10.2.6

If $v \perp w$: (a) $\|v + w\|^2 = \|v\|^2 + \|w\|^2$ (Pythagoras);
(b) $\|v + w\| = \|v - w\|$.

Solution. In the expansion of Exercise 10.2.4 the cross terms $\langle v,w\rangle + \langle w,v\rangle$ vanish. □

Exercise 10.2.7

If $w \neq 0$, then $(v - \frac{\langle v,w\rangle}{\langle w,w\rangle} w) \perp w$ (compare Example 9.4.1).

Solution. $\langle v,w\rangle - \frac{\langle v,w\rangle}{\langle w,w\rangle}\langle w,w\rangle = 0$. □

Observation 10.2.8

(a) (Cauchy–Schwarz) $|\langle v,w\rangle| \leq \|v\|\|w\|$, with equality iff $\{v,w\}$ is linearly dependent.
(b) (Triangle inequality) $\|v+w\| \leq \|v\| + \|w\|$.

Proof. (a) Trivial if $w = 0$. Otherwise, with $a = \frac{\langle v,w\rangle}{\langle w,w\rangle}$, Exercise 10.2.7 and Pythagoras give $\|v\|^2 = \|v - aw\|^2 + \|aw\|^2 \geq |a|^2\|w\|^2$, i.e. $|\langle v,w\rangle| \leq \|v\|\|w\|$; equality iff $v = aw$.

(b) Using $z+\bar{z} = 2\operatorname{Re} z \leq 2|z|$, $\|v+w\|^2 = \|v\|^2 + \langle v,w\rangle + \overline{\langle v,w\rangle} + \|w\|^2 = \|v\|^2 + 2\operatorname{Re}\langle v,w\rangle + \|w\|^2 \leq \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 = (\|v\| + \|w\|)^2$. ■

So $\|v-w\|$ is a distance on V:

Exercise 10.2.9

(a) $\|v - v\| = 0$.
(b) $\|v - w\| = \|w - v\|$.
(c) $\|v - w\| \leq \|v - u\| + \|u - w\|$.

Solution. As in Exercise 9.1.8, using Observation 10.2.8(b). □

Definition 10.2.10 (Orthogonal and orthonormal sets)

(a) Nonzero vectors $\{v_i\}_{i\in I}$ with $v_i \perp v_j$ for $i \neq j$ form an **orthogonal subset**; if all are unit vectors, an **orthonormal subset**.
(b) For $W \leq V$, a basis $\mathfrak{B}$ of W is an **orthogonal basis** if it is an orthogonal subset, an **orthonormal basis** if it is orthonormal.

Exercise 10.2.11

An orthogonal subset $\{v_i\}_{i\in I}$ is linearly independent.

Solution. If $\sum_{\text{finite}} c_i v_i = 0$, take the inner product with v_j: $c_j\|v_j\|^2 = 0$. □

Exercise 10.2.12

A basis $\{v_i\}$ of $W \leq V$ is orthonormal iff $\langle v_i, v_j\rangle = \delta_{ij}$.

Solution. This restates the definition. □

Exercise 10.2.13

(a) $\{(0,-i)^t, (i,0)^t\}$ is an orthonormal basis of $\mathbb{C}^2$ (dot product).
(b) $\{\frac{1}{\sqrt{3}}(1+i,i)^t, \frac{1}{\sqrt{6}}(1-i,-2)^t\}$ is an orthonormal basis of $\mathbb{C}^2$.

(c) $\{\frac{1}{\sqrt{2}}(1,0)^t, \frac{1}{\sqrt{6}}(1,2)^t\}$ is an orthonormal basis of $\mathbb{R}^2$ with the inner product of Exercise 10.1.9.
(d) $\{1, \sqrt{12}(t-\frac{1}{2})\}$ is an orthonormal basis of $\mathcal{P}_1(\mathbb{R})$ with the inner product of Exercise 10.1.11.
(e) For $f_n(x) = e^{inx} = \cos nx + i \sin nx$ $(n \in \mathbb{Z})$, $\{f_n\}_{n\in\mathbb{Z}}$ is an orthonormal subset of $C^0[0, 2\pi]$ (Exercise 10.1.13).

Solution. (a) Norms 1, and $0 \cdot \bar{i} + (-i) \cdot 0 = 0$. (b) $|1+i|^2 + |i|^2 = 3$, $|1-i|^2 + 4 = 6$, and $(1+i)\overline{(1-i)} + i \cdot \overline{(-2)} = (1+i)^2 - 2i = 0$. (c) $\langle (1,0),(1,0)\rangle = 2$, $\langle (1,2),(1,2)\rangle = 2-2-2+8 = 6$, $\langle (1,0),(1,2)\rangle = 2-0-2+0 = 0$. (d) $\int_0^1 1 = 1$, $\int_0^1 12(t-\frac{1}{2})^2 dt = 1$, $\int_0^1 (t-\frac{1}{2})dt = 0$. (e) $\frac{1}{2\pi}\int_0^{2\pi} e^{i(m-n)x}dx = \delta_{mn}$, since for $m \neq n$ the antiderivative $\frac{e^{i(m-n)x}}{i(m-n)}$ is 2π-periodic. □

Question 10.2.14

If V is a (finite-dimensional) inner product space and $W \leq V$, does W always have an orthonormal basis?

Definition 10.2.15 (Orthogonal complement)

$S^\perp = \{v \in V : w \perp v \text{ for all } w \in S\}$; for $W \leq V$, $W^\perp$ is the **orthogonal complement**.

Exercise 10.2.16

For $S \subseteq V$, $W \leq V$ with basis $\mathfrak{C}$: (a) $S^\perp \leq V$;
(b) $S^\perp = \langle S\rangle^\perp$;
(c) $W^\perp = \mathfrak{C}^\perp$.

Solution. As in Observation 9.1.15; in (a) use conjugate linearity: $\langle w, cv\rangle = \bar{c}\langle w, v\rangle = 0$. □

Exercise 10.2.17

For a basis $\mathfrak{B}$ of V and $W \leq V$: (a) $0^\perp = V$, $V^\perp = \mathfrak{B}^\perp = 0$;
(b) $W \cap W^\perp = 0$;
(c) $W \leq (W^\perp)^\perp$.

Solution. As in Exercise 9.1.16, using Exercise 10.2.1. □

Question 10.2.18

If V is finite-dimensional and $W \leq V$, is $\dim W^\perp = \dim V - \dim W$ and $V = W \oplus W^\perp$?

A **rigid motion** of V is $M : V \to V$ with $\|M(v) - M(w)\| = \|v - w\|$. For $F = \mathbb{C}$, Theorem 9.2.11 fails, since Observation 9.2.8(c) does.

Exercise 10.2.19

(a) On $\mathbb{C} = \mathbb{C}^1$ with the dot product, $M(\alpha) = \bar{\alpha}$ is a non-linear rigid motion fixing 0 (M is conjugate linear).
(b) On $\mathbb{C}^2$, $M((\alpha,\beta)^t) = (\bar{\alpha},\beta)^t$ is a non-linear rigid motion fixing 0, not even conjugate linear.

Solution. (a) $|\bar{\alpha}-\bar{\beta}| = |\overline{\alpha-\beta}| = |\alpha-\beta|$, but $M(i\cdot 1) = -i \neq iM(1)$. (b) $\|M(v)-M(w)\|^2 = |\overline{\alpha-\alpha'}|^2 + |\beta-\beta'|^2 = \|v-w\|^2$; $M(i(1,0)^t) = (-i,0)^t \neq iM((1,0)^t)$, and $M(i(0,1)^t) = (0,i)^t \neq \bar{i}M((0,1)^t)$. □

Remark 10.2.20

The $\mathbb{R}$-isomorphism $\gamma(a+ib) = (a,b)^t$, $\mathbb{C} \to \mathbb{R}^2$, preserves distance, so rigid motions of $\mathbb{C}$ and of $\mathbb{R}^2$ coincide. Conjugation is a reflection of $\mathbb{R}^2$ ($\mathbb{R}$-linear, not $\mathbb{C}$-linear); a $\mathbb{C}$-linear rigid motion of $\mathbb{C}$ (multiplication by $e^{i\theta}$) is a rotation of $\mathbb{R}^2$.

10.3 Gram–Schmidt orthogonalization

Motivation. Subtracting projections onto earlier vectors orthogonalizes any independent list, answering Questions 9.1.19, 9.4.8, 10.2.14, 10.2.18.

Theorem 10.3.1 (Gram–Schmidt orthogonalization)

Let V be an inner product space and $\{v_1,\dots,v_r\}$ linearly independent. (a) Put $w_1 = v_1$ and, for $2 \le i \le r$,

$$w_i = v_i - \frac{\langle v_i, w_{i-1}\rangle}{\langle w_{i-1}, w_{i-1}\rangle} w_{i-1} - \cdots - \frac{\langle v_i, w_1\rangle}{\langle w_1, w_1\rangle} w_1.$$

Then $\langle v_1,\dots,v_r\rangle = \langle w_1,\dots,w_r\rangle$ and $\{w_1,\dots,w_r\}$ is an orthogonal subset.
(b) Hence $\{\frac{1}{\|w_1\|}w_1,\dots,\frac{1}{\|w_r\|}w_r\}$ is an orthonormal subset.
(c) In particular, if $r = \dim V$, it is an orthonormal basis of V.

Proof. Induction on i: let $\langle v_1,\dots,v_i\rangle = \langle w_1,\dots,w_i\rangle$ with $\{w_1,\dots,w_i\}$ orthogonal. Then $w_{i+1} \in v_{i+1} + \langle w_1,\dots,w_i\rangle$ gives equal spans at $i+1$; $w_{i+1} \neq 0$ since $v_{i+1} \notin \langle v_1,\dots,v_i\rangle$; and for $j \le i$, $\langle w_{i+1}, w_j\rangle = \langle v_{i+1}, w_j\rangle - \frac{\langle v_{i+1},w_j\rangle}{\langle w_j,w_j\rangle}\langle w_j, w_j\rangle = 0$. Normalize for (b), (c) (Exercise 10.2.11). ■

Exercise 10.3.2

Apply Gram–Schmidt to the basis $\mathfrak{B}$: (a) $\mathfrak{B} = \{(1,1,0)^t,(-1,0,2)^t,(2,1,1)^t\}$ of $\mathbb{R}^3$ (dot product); (b) $\mathfrak{B} = \{(3,1,0,1)^t,(0,-1,2,0)^t,(1,1,-1,-1)^t\}$ of a subspace W of $\mathbb{R}^4$; (c) $\mathfrak{B} = \{(1,1)^t,(0,1)^t\}$ of $\mathbb{R}^2$ with the inner product of Exercise 10.1.9; (d) $\mathfrak{B} = \{1,t,t^2\}$ of $\mathcal{P}_2(\mathbb{R})$ with the inner product of Exercise 10.1.11.

Solution. (a) $w_1 = (1,1,0)$, $w_2 = (-\frac{1}{2},\frac{1}{2},2)$, $w_3 = (\frac{2}{3},-\frac{2}{3},\frac{1}{3})$; orthonormal basis $\frac{1}{\sqrt{2}}(1,1,0)$, $\frac{1}{3\sqrt{2}}(-1,1,4)$, $\frac{1}{3}(2,-2,1)$.

(b) $w_1 = (3,1,0,1)$, $w_2 = \frac{1}{11}(3,-10,22,1)$, $w_3 = \frac{1}{9}(3,2,1,-11)$; orthonormal basis $\frac{1}{\sqrt{11}}(3,1,0,1)$, $\frac{1}{\sqrt{594}}(3,-10,22,1)$, $\frac{1}{\sqrt{135}}(3,2,1,-11)$.

(c) $\langle(1,1),(1,1)\rangle = 2$, $\langle(0,1),(1,1)\rangle = 1$, so $w_2 = (0,1) - \frac{1}{2}(1,1) = (-\frac{1}{2},\frac{1}{2})$ with $\langle w_2,w_2\rangle = \frac{3}{2}$; orthonormal basis $\frac{1}{\sqrt{2}}(1,1)$, $\frac{1}{\sqrt{6}}(-1,1)$.

(d) 1, $t-\frac{1}{2}$, $t^2-t+\frac{1}{6}$; orthonormal basis 1, $\sqrt{3}(2t-1)$, $\sqrt{5}(6t^2-6t+1)$. □

Exercise 10.3.3

If $\{v_1,\dots,v_r\}$ in Theorem 10.3.1 is already orthogonal, then $w_i = v_i$ for all i.

Solution. All the subtracted terms have $\langle v_i, w_j\rangle = \langle v_i, v_j\rangle = 0$, by induction. □

Corollary 10.3.4 (Perp theorem, inner product case)

If V is a finite-dimensional inner product space and $W \leq V$, then $\dim V = \dim W + \dim W^\perp$.

Proof. Extend an orthonormal basis $\{w_1, \ldots, w_r\}$ of W to a basis of V and apply Gram–Schmidt: by Exercise 10.3.3 this gives an orthonormal basis $\{w_1, \ldots, w_n\}$ of V. Then $W^\perp = \langle w_{r+1}, \ldots, w_n\rangle$, since $v = \sum a_i w_i \in W^\perp$ iff $a_j = \langle v, w_j\rangle = 0$ for $j \leq r$. ■

Corollary 10.3.5

If V is a finite-dimensional inner product space and $W \leq V$, then $V = W \oplus W^\perp$.

Proof. $W \cap W^\perp = 0$ (Exercise 10.2.17) and the dimensions add up (Corollary 10.3.4). ■

Exercise 10.3.6

For a finite-dimensional inner product space V and $U, W \leq V$: (a) $W = (W^\perp)^\perp$;
(b) $(U + W)^\perp = U^\perp \cap W^\perp$;
(c) $(U \cap W)^\perp = U^\perp + W^\perp$.

Solution. (a) $W \leq (W^\perp)^\perp$ and both have dimension $\dim W$ by Corollary 10.3.4 (applied twice). (b) $v \perp U + W$ iff $v \perp U$ and $v \perp W$. (c) Apply (b) to $U^\perp, W^\perp$: $(U^\perp + W^\perp)^\perp = U^{\perp\perp} \cap W^{\perp\perp} = U \cap W$, and take $\perp$ using (a). □

Remark 10.3.7

Cancelling $W^\perp$ from $W \oplus W^\perp = V = W^\perp \oplus (W^\perp)^\perp$ to get (a) is nonsense: complements in a direct sum are not unique (Remark 7.6.10).

With the dot product on $\mathbb{R}^n$, $AX = 0$ reads $\langle [A]_i^t, X\rangle = 0$ for all rows i, so

$$\ker L_A = \langle [A]_1^t, \ldots, [A]_m^t\rangle^\perp = [\text{row space of } A]^\perp.$$

Example 10.3.8

For $W = \langle X_1, \ldots, X_m\rangle \leq \mathbb{R}^n$, let A have rows X_i^t. Then $W^\perp = \ker L_A$ and $\dim W =$ row rank $=$ column rank $= \dim \operatorname{im} L_A$, so the dimension theorem gives $\dim W^\perp = n - \dim W$.

Proposition 10.3.9 (The three pillars, $F = \mathbb{R}$)

Of (1) the dimension theorem (for L_A), (2) the rank theorem, (3) the perp theorem (for the dot product on $\mathbb{R}^n$), any two imply the third.

Proof. With W the row space, $\dim \ker L_A = \dim W^\perp$, and the three are the linear relations (1) $\dim \ker L_A + \dim \operatorname{im} L_A = n$, (2) $\dim W = \dim \operatorname{im} L_A$, (3) $\dim W + \dim W^\perp = n$. ■

This gives a proof of the rank theorem without row reduction. Over $\mathbb{C}$, X^tY is not an inner product, so we need:

Exercise 10.3.10

For $W \leq \mathbb{C}^n$, $\overline{W} = \{\bar{X} : X \in W\}$ is a subspace of $\mathbb{C}^n$ with $\dim \overline{W} = \dim W$.

Solution. $\overline{W}$ is closed under sums and under $c\bar{X} = \overline{\bar{c}X}$. If $\{X_i\}$ is a basis of W, then $\{\bar{X}_i\}$ is a basis of $\overline{W}$ (conjugate any relation). □

Proposition 10.3.11 (The three pillars)

For $F = \mathbb{R}$ or $\mathbb{C}$, of (1) the dimension theorem (for L_A), (2) the rank theorem, (3) the perp theorem (for the dot product on F^n), any two imply the third.

Proof. As in Proposition 10.3.9, using $A\bar{X} = 0 \iff \langle [A]_i^t, X\rangle = 0$: the solution space of $A\bar{X} = 0$ is $W^\perp$, and it equals $\overline{\ker L_A}$, of dimension $\dim \ker L_A$ (Exercise 10.3.10). ■

10.4 Standard basis versus orthonormal basis

Motivation. In any orthonormal basis, coordinates are inner products and the inner product is the dot product of coordinates.

For the dot product on F^n and the standard basis $\mathcal{E}$: (a) $X = \sum \langle X, e_i\rangle e_i$;

(b) $\langle X, Y\rangle = [X]_{\mathcal{E}}^t \overline{[Y]_{\mathcal{E}}}$.

Observation 10.4.1

If $\mathfrak{B} = \{v_1, \ldots, v_n\}$ is an orthogonal basis of V:
(a)′ $v = \sum_i \frac{\langle v, v_i\rangle}{\langle v_i, v_i\rangle} v_i$. If $\mathfrak{B}$ is an orthonormal basis:
(a) $v = \sum_i \langle v, v_i\rangle v_i$, i.e. the i-th coordinate of $[v]_{\mathfrak{B}}$ is $\langle v, v_i\rangle$;
(b) $\langle v, w\rangle = [v]_{\mathfrak{B}}^t \cdot \overline{[w]_{\mathfrak{B}}}$.

Proof. (a) If $v = \sum a_i v_i$, then $\langle v, v_j\rangle = a_j \langle v_j, v_j\rangle$; this gives (a)′ and (a).

(b) For $w = \sum b_i v_i$, $\langle v, w\rangle = \sum_{i,j} a_i \bar{b}_j \langle v_i, v_j\rangle = \sum a_i \bar{b}_i$. ■

E.g. the foot of the perpendicular from $X = (1, 2, 3)^t$ to the xy-plane is $(1, 2, 0)^t = \langle X, e_1\rangle e_1 + \langle X, e_2\rangle e_2$.

Observation 10.4.2

Let W be a finite-dimensional subspace of an inner product space V with orthonormal basis $\{v_1, \ldots, v_m\}$. (a) $V = W \oplus W^\perp$.
(b) (Closest vector problem) For $v \in V$, $w = \langle v, v_1\rangle v_1 + \cdots + \langle v, v_m\rangle v_m \in W$ satisfies $v - w \in W^\perp$, and w is the unique vector of W closest to v.

Proof. (a) With w as in (b), $\langle v - w, v_i\rangle = \langle v, v_i\rangle - \langle v, v_i\rangle = 0$ for all i, so $v - w \in W^\perp$ and $V = W + W^\perp$; also $W \cap W^\perp = 0$.

(b) For $u \in W$, $w - u \in W$ is perpendicular to $v - w$, so by Pythagoras $\|v - u\|^2 = \|v - w\|^2 + \|w - u\|^2 \geq \|v - w\|^2$, with equality iff $u = w$. ■

Exercise 10.4.3

(a) In Euclidean $\mathbb{R}^3$, find an orthonormal basis of $W = \langle (1,1,2)^t \rangle^\perp$ and the vector of W closest to $(1,0,-1)^t$.
(b) Find the foot of the perpendicular from $(1,0,0,0)^t$ to the subspace W of Exercise 10.3.2(b).

Solution. (a) $\frac{1}{\sqrt{2}}(1,-1,0)$, $\frac{1}{\sqrt{3}}(1,1,-1)$. The closest vector is $X - \frac{\langle X,Y\rangle}{\langle Y,Y\rangle}Y = (1,0,-1) + \frac{1}{6}(1,1,2) = (\frac{7}{6}, \frac{1}{6}, -\frac{2}{3})$ (with $Y = (1,1,2)$).
(b) Summing $\langle v, u_i\rangle u_i$ over the orthonormal basis found there: $(\frac{9}{10}, \frac{4}{15}, \frac{2}{15}, \frac{1}{30})$. □

Exercise 10.4.4

If $\{v_1, \ldots, v_n\}$ is an orthonormal basis and $v = \sum a_i v_i$, then $\|v\|^2 = \sum |a_i|^2 = \sum |\langle v, v_i\rangle|^2$.

Solution. Observation 10.4.1(b) with $w = v$, and (a). □

Observation 10.4.5 (Bessel's inequality)

If $\{v_1, \ldots, v_m\}$ is an orthonormal subset of an inner product space V (possibly infinite-dimensional), then $\sum_{i=1}^m |\langle v, v_i\rangle|^2 \le \|v\|^2$ for all $v \in V$.

Proof. With $W = \langle v_1, \ldots, v_m\rangle$, Observation 10.4.2 gives $v = w + w'$, $w = \sum \langle v, v_i\rangle v_i \in W$, $w' \in W^\perp$, and by Pythagoras $\|v\|^2 = \|w\|^2 + \|w'\|^2 \ge \|w\|^2 = \sum |\langle v, v_i\rangle|^2$. ■

For an orthonormal subset $\{v_i\}$, $\langle v, v_i\rangle$ is the i-th **Fourier coefficient** of v.

Exercise 10.4.6

With $f_n(x) = e^{inx}$ in $C^0[0, 2\pi]$ (Exercise 10.2.13(e)) and $f(x) = x$: (a) $\langle f, f_0\rangle = \pi$ and $\langle f, f_n\rangle = \frac{i}{n}$ for $n \neq 0$;
(b) $\|f\|^2 = \frac{4}{3}\pi^2$;
(c) Bessel's inequality for $\{f_n\}_{-k \le n \le k}$ gives $\sum_{n=1}^k \frac{1}{n^2} \le \frac{\pi^2}{6}$, hence $\sum_{n=1}^\infty \frac{1}{n^2} \le \frac{\pi^2}{6}$.

Solution. (a) $\frac{1}{2\pi}\int_0^{2\pi} x\,dx = \pi$; for $n \neq 0$, integrating by parts, $\frac{1}{2\pi}\int_0^{2\pi} xe^{-inx}dx = \frac{1}{2\pi}\left[\frac{xe^{-inx}}{-in}\right]_0^{2\pi} = \frac{1}{-in} = \frac{i}{n}$. (b) $\frac{1}{2\pi}\int_0^{2\pi} x^2 dx = \frac{4\pi^2}{3}$. (c) $\frac{4}{3}\pi^2 \ge \pi^2 + 2\sum_{n=1}^k \frac{1}{n^2}$. □

10.5 Isomorphisms of inner product spaces

Motivation. A finite-dimensional inner product space is F^n with the dot product under another name.

Definition 10.5.1 (Isomorphism of inner product spaces)

Inner product spaces V, W over F are **isomorphic** if there is a vector space isomorphism $\varphi : V \to W$ with $\langle \varphi(u), \varphi(v)\rangle = \langle u, v\rangle$ for all $u, v \in V$; φ is an **inner product space isomorphism**. Equivalently, with $(\varphi \times \varphi)(u, v) = (\varphi(u), \varphi(v))$, the square

$$\begin{array}{ccc} V \times V & \xrightarrow{\langle\,,\rangle} & F \\ {\scriptstyle \varphi\times\varphi}\downarrow & & \downarrow{\scriptstyle I} \\ W \times W & \xrightarrow[\langle\,,\rangle]{} & F \end{array}$$

commutes.

Exercise 10.5.2

Is $\varphi \times \varphi : V \times V \to W \times W$ a vector space isomorphism between the product spaces?

Solution. Yes: it is linear componentwise, with inverse $\varphi^{-1} \times \varphi^{-1}$. □

Exercise 10.5.3

Isomorphism of inner product spaces is an equivalence relation.

Solution. I_V works; inverses of isomorphisms preserve inner products ($\langle \varphi^{-1}u, \varphi^{-1}v\rangle = \langle \varphi\varphi^{-1}u, \varphi\varphi^{-1}v\rangle$); composites of isomorphisms preserve them. □

Example 10.5.4

For an orthonormal basis $\mathfrak{B} = \{v_1, \ldots, v_n\}$ of V and the coordinate isomorphism $\alpha_{\mathfrak{B}}(v_i) = e_i$, Observation 10.4.1(b) gives $\langle \alpha_{\mathfrak{B}}(v), \alpha_{\mathfrak{B}}(w)\rangle = \sum a_i \bar{b}_i = \langle v, w\rangle$ for $v = \sum a_i v_i$, $w = \sum b_i v_i$. So $\alpha_{\mathfrak{B}}$ is an inner product space isomorphism onto F^n with the dot product.

Theorem 10.5.5 (Classification of finite-dimensional inner product spaces)

For finite-dimensional inner product spaces V, W over F the following are equivalent: (1) V and W are isomorphic as inner product spaces; (2) $\dim V = \dim W$.

Proof. (1)⇒(2): clear. (2)⇒(1): for orthonormal bases $\mathfrak{B}, \mathfrak{C}$, $(\alpha_{\mathfrak{C}})^{-1} \circ \alpha_{\mathfrak{B}}$ is an inner product space isomorphism (Example 10.5.4). ■

Exercise 10.5.6

Write explicit inner product space isomorphisms (a) from $\mathbb{R}^2$ with the dot product to $\mathbb{R}^2$ with the inner product of Exercise 10.1.9; (b) from F^4 with the dot product to $\mathfrak{M}_{2,2}(F)$ with the trace form of Exercise 10.1.14.

Solution. (a) Send e_1, e_2 to the orthonormal basis of Exercise 10.2.13(c): $\varphi(a, b) = \frac{a}{\sqrt{2}}(1, 0) + \frac{b}{\sqrt{6}}(1, 2)$. (b) The matrix units $E_{11}, E_{12}, E_{21}, E_{22}$ are orthonormal for $\operatorname{tr}(A^t \bar{B})$; send $e_1, \ldots, e_4$ to them, i.e. $(a, b, c, d)^t \mapsto \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. □

For $F = \mathbb{R}$ the results of §9.2 on rigid motions carry over:

Theorem 10.5.7

Let V be a finite-dimensional inner product space over $\mathbb{R}$. (a) A rigid motion L of V with $L(0) = 0$ is linear. (b) Every rigid motion of V is a translation composed with a linear rigid motion. (c) Every rigid motion of V is a bijection.

First proof. V is isomorphic as an inner product space to $\mathbb{R}^n$ with the dot product (Theorem 10.5.5), and the statements are Theorem 9.2.11 and Corollary 9.2.12. ■

Second proof. The second proof of Theorem 9.2.11(a) works verbatim with an orthonormal basis in place of $\mathcal{E}$; (b), (c) follow as before. ■

Exercise 10.5.8

Let $\gamma : \mathbb{C}^n \to \mathbb{R}^{2n}$, $\gamma((a_1 + ib_1, \dots, a_n + ib_n)^t) = (a_1, b_1, \dots, a_n, b_n)^t$, with the dot products on both spaces. (a) γ is an isomorphism of $\mathbb{R}$-vector spaces.
(b) $\|\gamma(X) - \gamma(Y)\| = \|X - Y\|$.
(c) If M is a rigid motion of $\mathbb{C}^n$, then $\gamma \circ M \circ \gamma^{-1}$ is a rigid motion of $\mathbb{R}^{2n}$; hence M is a bijection.

Solution. (a) Clear.

(b) $|a + ib|^2 = a^2 + b^2$ coordinatewise.

(c) $\gamma M \gamma^{-1}$ preserves distances, so is a bijection (Corollary 9.2.12), and so is $M = \gamma^{-1}(\gamma M \gamma^{-1})\gamma$. □

10.6 Orthogonal groups and unitary groups

Motivation. Linear isometries of real and complex spaces form $\mathbf{O}(V)$ and $\mathbf{U}(V)$; their matrices satisfy $A^*A = I$.

Spaces are finite-dimensional.

Definition 10.6.1 (Orthogonal and unitary groups)

(a) For a real inner product space V: $\mathbf{O}(V) = \mathbf{O}(V, \langle\cdot,\cdot\rangle) = \{L \in \mathfrak{L}(V, V) : \|Lv - Lw\| = \|v - w\| \ \forall v, w\}$, the **orthogonal group**; its elements are **orthogonal operators**.
(b) For a complex inner product space V: $\mathbf{U}(V) = \mathbf{U}(V, \langle\cdot,\cdot\rangle)$ is defined the same way, the **unitary group**; its elements are **unitary operators**.

They depend on the inner product; their elements are injective, hence invertible.

Exercise 10.6.2

For $G = \mathbf{O}(V)$ or $\mathbf{U}(V)$: (a) $I_V \in G$;
(b) $L, M \in G \Rightarrow L \circ M \in G$;
(c) $L \in G \Rightarrow L^{-1} \in G$.

Solution. As in Exercise 9.2.7. □

Definition 10.6.3 (Unitary matrices)

With the Hermitian dot product on $\mathbb{C}^n$, $\mathbf{U}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{C}) : L_A \in \mathbf{U}(\mathbb{C}^n)\} = \{[L]_{\mathcal{E}}^{\mathcal{E}} : L \in \mathbf{U}(\mathbb{C}^n)\}$, the (complex) **unitary group**; its elements are **unitary matrices**. Likewise $\mathbf{O}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{R}) : L_A \in \mathbf{O}(\mathbb{R}^n)\}$.

Remark 10.6.4

For an orthonormal basis $\mathfrak{B}$ of V, $\{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{U}(V)\} = \mathbf{U}(n)$ (similarly for $\mathbf{O}$); cf. Exercises 10.6.11, 9.6.11 and §13.3.

Observation 10.6.5

For an n-dimensional inner product space V and $L \in \mathfrak{L}(V,V)$ the following are equivalent: (1) $\|Lv - Lw\| = \|v - w\|$ (i.e. $L \in \mathbf{O}(V)$ or $\mathbf{U}(V)$); (2) $\|Lv\| = \|v\|$; (3) $\langle Lv, Lv\rangle = \langle v, v\rangle$; (4) $\langle Lv, Lw\rangle = \langle v, w\rangle$; (5) if $\mathfrak{B}$ is an orthonormal basis, so is $L(\mathfrak{B})$.

Proof. As in Observation 9.3.4 (for (5)⇒(2) use Exercise 10.4.4), except (3)⇒(4) for $F = \mathbb{C}$: expanding (3) at $v + w$ and at $v + iw$ gives $\mathrm{Re}\langle Lv, Lw\rangle = \mathrm{Re}\langle v, w\rangle$ and $-i\langle Lv, Lw\rangle + i\langle Lw, Lv\rangle = -i\langle v, w\rangle + i\langle w, v\rangle$, i.e. $\mathrm{Im}\langle Lv, Lw\rangle = \mathrm{Im}\langle v, w\rangle$. ■

Exercise 10.6.6

One may add to Observation 10.6.5: (2)′ $\|v\| = 1 \Rightarrow \|Lv\| = 1$; (5)′ some orthonormal basis $\mathfrak{B}$ has $L\mathfrak{B}$ orthonormal.

Solution. As in Exercise 9.3.5. □

Definition 10.6.7 (Adjoint matrix)

For $A \in \mathfrak{M}_{m,n}(F)$, $A^* = \overline{A^t} = \bar{A}^t$ is the **adjoint matrix** (for real A, $A^* = A^t$; unrelated to the classical adjoint of §6.7).

Exercise 10.6.8

For $A, B \in \mathfrak{M}_{m,n}(F)$, $C \in \mathfrak{M}_{n,r}(F)$, $c \in F$: (a) $(A + B)^* = A^* + B^*$;
(b) $(cA)^* = \bar{c}A^*$;
(c) $(A^*)^* = A$;
(d) $(AC)^* = C^*A^*$.

Solution. Conjugation and transposition are additive, conjugation is multiplicative, $(AC)^t = C^tA^t$, and both operations are involutions. □

Exercise 10.6.9

For $A \in \mathfrak{M}_{n,n}(\mathbb{C})$ (Hermitian dot product on $\mathbb{C}^n$) the following are equivalent: (1) $\|AX - AY\| = \|X - Y\|$ (i.e. $A \in \mathbf{U}(n)$); (2) $\|AX\| = \|X\|$; (3) $\langle AX, AX\rangle = \langle X, X\rangle$; (4) $\langle AX, AY\rangle = \langle X, Y\rangle$; (5) the columns form an orthonormal basis of $\mathbb{C}^n$; (6) $A^t\bar{A} = I = \bar{A}A^t$, i.e. $A^*A = I = AA^*$, $A^{-1} = A^*$; (7) $A^t \in \mathbf{U}(n)$; (7)′ $\bar{A} \in \mathbf{U}(n)$; (7)″ $A^* \in \mathbf{U}(n)$; (8) the rows form an orthonormal basis of $\mathfrak{M}_{1,n}(\mathbb{C})$.

Solution. (1)–(5) is Observation 10.6.5 with $\mathfrak{B} = \mathcal{E}$. The (i,j) entry of A^*A is $\overline{\langle [A]^i, [A]^j\rangle}$, which gives (5)⇔(6). (6) is preserved by $A \mapsto A^t, \bar{A}, A^*$ (take transposes and conjugates of $A^*A = I = AA^*$), giving (7), (7)′, (7)″. (8) is (5) for A^t. □

Exercise 10.6.10

For $G = \mathbf{O}(n)$ or $\mathbf{U}(n)$: (a) $I_n \in G$;
(b) $A, B \in G \Rightarrow AB \in G$;
(c) $A \in G \Rightarrow A^{-1} \in G$.

Solution. $(AB)^*AB = B^*A^*AB = I$; $(A^{-1})^*A^{-1} = (A^*)^*A^* = AA^* = I$. □

So $\mathbf{U}(n) = \{A \in \mathfrak{M}_{n,n}(\mathbb{C}) : A^*A = I\}$; defining L^* for $\mathbf{U}(V)$ is Question 10.7.2.

Exercise 10.6.11

If $\mathfrak{B}, \mathfrak{C}$ are orthonormal bases of an n-dimensional complex inner product space V, then $[I]_{\mathfrak{C}}^{\mathfrak{B}} \in \mathbf{U}(n)$.

Solution. The columns are $[w_j]_{\mathfrak{B}}$, $w_j \in \mathfrak{C}$, and $\langle [w_i]_{\mathfrak{B}}, [w_j]_{\mathfrak{B}} \rangle = \langle w_i, w_j \rangle = \delta_{ij}$ (Observation 10.4.1(b)). □

Exercise 10.6.12

If $A \in \mathbf{U}(n)$, then $|\det A| = 1$.

Solution. $1 = \det(A^*A) = \overline{\det A} \det A = |\det A|^2$. □

Definition 10.6.13 (Special unitary group)

$\mathbf{SU}(n) = \{A \in \mathbf{U}(n) : \det A = 1\}$.

Exercise 10.6.14

(a) $\mathbf{SU}(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} : \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$. What is $\mathbf{U}(2)$?
(b) Are the elements of $\mathbf{SU}(2)$ diagonalizable over $\mathbb{C}$? Those of $\mathbf{U}(2)$?

Solution. (a) For $A = \left(\begin{smallmatrix} \alpha & z \\ \beta & w \end{smallmatrix}\right) \in \mathbf{SU}(2)$, $|\alpha|^2 + |\beta|^2 = 1$ and $(z, w)^t$ is a unit vector in $\langle (\alpha, \beta)^t \rangle^{\perp} = \langle (-\bar{\beta}, \bar{\alpha})^t \rangle$: $(z, w) = \lambda(-\bar{\beta}, \bar{\alpha})$, $|\lambda| = 1$, and $\det A = \lambda = 1$. The converse is clear. Hence $\mathbf{U}(2) = \left\{ \left(\begin{smallmatrix} \alpha & -\lambda\bar{\beta} \\ \beta & \lambda\bar{\alpha} \end{smallmatrix}\right) : |\alpha|^2 + |\beta|^2 = 1,\ |\lambda| = 1 \right\}$.

(b) Yes in both cases (normal matrices, Chapter 15). Directly: take a unit eigenvector v_1 of unitary A and an orthonormal basis $\{v_1, v_2\}$; $Av_2 \perp Av_1 = \lambda v_1$ gives $Av_2 \in \langle v_2 \rangle$. □

10.7 The adjoint matrix and its applications

Motivation. The identity $\langle AX, Y \rangle = \langle X, A^*Y \rangle$ yields least squares and minimal solutions.

Observation 10.7.1

With dot products on F^n, F^m and $A \in \mathfrak{M}_{m,n}(F)$: $\langle AX, Y \rangle = \langle X, A^*Y \rangle$ and $\langle Y, AX \rangle = \langle A^*Y, X \rangle$ for $X \in F^n$, $Y \in F^m$.

Proof. $\langle AX, Y \rangle = (AX)^t\bar{Y} = X^tA^t\bar{Y}$, and since $\overline{A^*} = A^t$, $\langle X, A^*Y \rangle = X^t\overline{A^*Y} = X^tA^t\bar{Y}$. The second identity is the conjugate of the first. ■

Question 10.7.2

For finite-dimensional inner product spaces V, W and $L \in \mathfrak{L}(V,W)$, does there exist $L^* \in \mathfrak{L}(W,V)$ with $\langle Lv, w\rangle = \langle v, L^*w\rangle$ for all $v \in V$, $w \in W$? What does it mean?

For the rest of this section F^n carries the dot product.

Least squares approximation (LSA). An **LSA solution** of $AX = B$ ($A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$) minimizes $\|AX_0 - B\|$. E.g. fitting $g = \sum_{j\le n} c_j t^j$ to points $(a_i, b_i)_{i\le m}$, $m > n$, minimizes $\sum_i |g(a_i) - b_i|^2 = \|AX - B\|^2$ with $A = (a_i^j)$, $X = (c_j)$, $B = (b_i)$ ($m = n$: Lagrange interpolation).

Proposition 10.7.3

For $A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$: (a) an LSA solution X_0 ($\|AX_0 - B\| \le \|AX - B\|$ for all X) always exists; (b) if L_A is injective ($\operatorname{rk} A = n$), it is unique.

Proof. By Observation 10.4.2, $B = AX_0 + C$ with $C \in (\operatorname{im} A)^\perp$, and AX_0 is the unique closest vector of $\operatorname{im} A$ to B; if L_A is injective, it determines X_0. ■

Proposition 10.7.4

The solutions of $(A^*A)X = A^*B$ are exactly the LSA solutions.

Proof. X_0 is an LSA solution iff $AX_0 - B \in (\operatorname{im} A)^\perp$, i.e. $0 = \langle AX, AX_0 - B\rangle = \langle X, A^*(AX_0 - B)\rangle$ for all X (Observation 10.7.1), i.e. $A^*(AX_0 - B) = 0$. ■

Exercise 10.7.5

Find the least squares line for the points $(1,1)$, $(2,3)$, $(3,3)$, $(4,2)$ of $\mathbb{R}^2$.

Solution. For $y = ct + d$: $A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}$, $B = (1,3,3,2)^t$, $A^*A = \left(\begin{smallmatrix} 30 & 10 \\ 10 & 4 \end{smallmatrix}\right)$, $A^*B = (24, 9)^t$. Solving, $(c,d) = (\frac{3}{10}, \frac{3}{2})$: $y = 0.3t + 1.5$. □

Exercise 10.7.6

Solve the LSA problem for $x + 3y + z = 3$, $2x - y - 2z = 2$, $5x + y - 3z = 6$.

Solution. The coefficient matrix has rank 2, so the normal equations $(A^*A)X = A^*B$ have a line of solutions: $X = (\frac{47}{42}, \frac{4}{7}, 0)^t + t(5, -4, 7)^t$, $t \in \mathbb{R}$. □

Observation 10.7.7

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{rk}(A^*A) = \operatorname{rk} A$;
(b) if $m \ge n$ and $\operatorname{rk} A = n$, then $A^*A \in \mathfrak{M}_{n,n}(F)$ is invertible.

Proof. (a) By the dimension theorem it suffices that $\ker(A^*A) = \ker A$. If $A^*AX = 0$, then $0 = \langle X, A^*AX\rangle = \langle (A^*)^*X, AX\rangle = \langle AX, AX\rangle$ (Observation 10.7.1, Exercise 10.6.8(c)), so $AX = 0$. (b) follows. ■

Exercise 10.7.8

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{rk} A = \operatorname{rk} A^t = \operatorname{rk} \bar{A} = \operatorname{rk} A^*$;
(b) $\operatorname{rk}(AA^*) = \operatorname{rk} A$.

Solution. (a) Rank theorem and Exercise 10.3.10.

(b) Observation 10.7.7(a) for A^* and (a). □

Observation 10.7.9

For $A \in \mathfrak{M}_{m,n}(F)$: (a) $\operatorname{im}(A^*) = (\ker A)^\perp$, i.e. $\ker A = (\operatorname{im} A^*)^\perp$;
(b) $\operatorname{im} A = (\ker A^*)^\perp$, i.e. $\ker A^* = (\operatorname{im} A)^\perp$.

Proof. By Exercise 10.3.6(a) and $A \leftrightarrow A^*$ it suffices to show $\ker A = (\operatorname{im} A^*)^\perp$: $AX = 0$ gives $\langle X, A^*Y\rangle = \langle AX, Y\rangle = 0$; conversely $\langle Y, AX\rangle = \langle A^*Y, X\rangle = 0$ for all Y gives $AX = 0$. ■

Minimal solution problem. If $AX = B$ is solvable, find the solution of smallest norm (the foot of the perpendicular from 0 to the solution set).

Proposition 10.7.10

Let $AX = B$ ($A \in \mathfrak{M}_{m,n}(F)$, $B \in F^m$) have a solution. (a) There is a unique minimal solution X_0 ($AX_0 = B$ and $\|X_0\| \le \|X\|$ for every solution X).
(b) X_0 is the unique solution lying in $\operatorname{im}(A^*)$.
(c) If $(AA^*)Y_0 = B$, then $X_0 = A^*Y_0$.

Proof. (i) Let $AX_1 = B$. By Observation 10.7.9, $F^n = \operatorname{im}(A^*) \oplus \ker A$, so $X_1 = A^*Y_0 + Z$ with $Z \in \ker A$. Put $X_0 = A^*Y_0$; then $AX_0 = AX_1 = B$, and X_0 is a solution in $\operatorname{im} A^*$.

(ii) Every solution is $X = X_0 + Z'$ with $Z' \in \ker A = (\operatorname{im} A^*)^\perp$, so $X_0 \perp Z'$ and $\|X\|^2 = \|X_0\|^2 + \|Z'\|^2 \ge \|X_0\|^2$, with equality iff $Z' = 0$.

(iii) If $X \in \operatorname{im} A^*$ is a solution, then $X_0 - X \in \operatorname{im} A^* \cap \ker A = 0$. (c) follows: $A^*Y_0 \in \operatorname{im} A^*$ is a solution. ■

Exercise 10.7.11

Find the minimal solutions of (a) $x + 3y + z = 3$, $2x - y - 2z = 2$;
(b) $x + 3y + z = 3$, $2x - y - 2z = 2$, $5x + y - 3z = 7$.

Solution. (a) $AA^* = \begin{pmatrix} 11 & -3 \\ -3 & 9 \end{pmatrix}$, and $Y_0 = (AA^*)^{-1}(3,2)^t$ gives $X_0 = A^*Y_0 = (\frac{19}{18}, \frac{34}{45}, -\frac{29}{90})^t$. (b) Equation 3 = equation 1 $+2\times$ equation 2: same solutions, same X_0 as (a). □